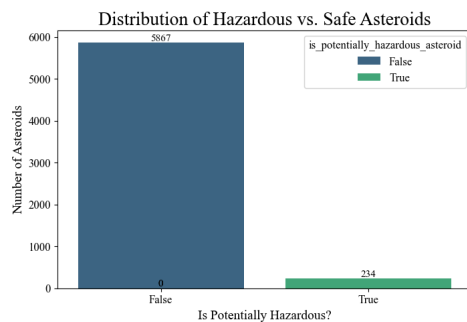


The Asteroid Watch: Securing Our Corporate Future

In the cold silence of our solar system, millions of rocks, remnants from the birth of our planets, are in constant motion. While most are harmless debris, a select few are on a potential collision course with Earth.

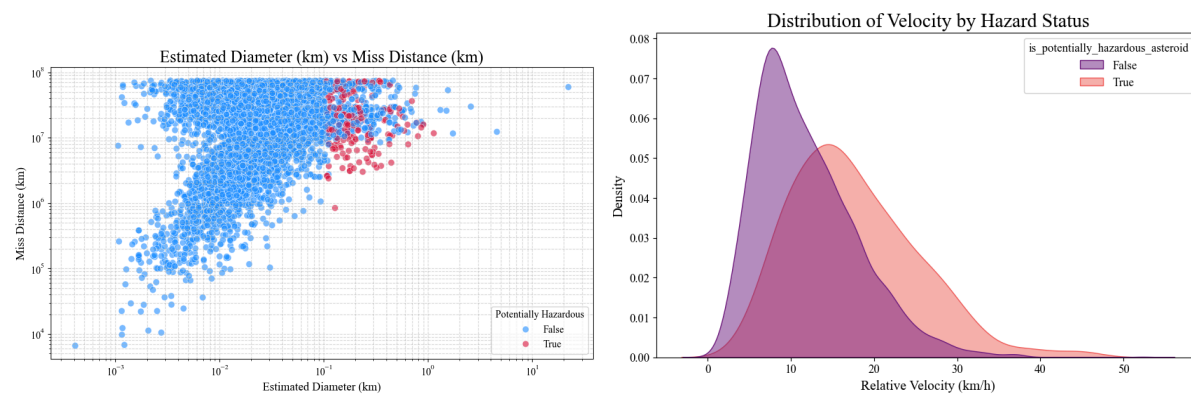
The Challenge: The Needle in the Haystack



Space is crowded with millions of objects. While most are harmless "space rocks," a tiny fraction are **Potentially Hazardous Asteroids (PHAs)**. The difficulty lies in the data: out of thousands of identified objects, only a small percentage (roughly 4%) are truly dangerous. For a leader, this represents the ultimate "Black Swan"—a low-probability, high-impact event that requires a plan.

The Strategy: Predictive Intelligence

To navigate this risk, we moved beyond simple observation. We implemented predictive models to categorize asteroids based on three critical metrics:



- **Proximity:** How close is the "near miss"?
- **Velocity:** How fast is the object traveling?
- **Magnitude:** What is the scale and potential impact force?

The Breakthrough: Precision Over Noise

Our analysis revealed a significant imbalance: we have plenty of data on safe asteroids, but "rare event" data on hazardous ones is scarce. To solve this, our team refined our models to ensure we weren't just "guessing," but were accurately identifying the physical characteristics—like diameter and orbital patterns—that signal a true threat. We've moved from reactive spotting to proactive identification.

The Bottom Line

To overcome this issue, we experimented with three different types of models to see which ones are the most reliable and accurate in this scenario. We understand that this is a preliminary step into predicting a hazardous asteroid. By gathering one year's worth of data, we were able to train the model to predict patterns rather than just simply guessing.

The sky is no longer a mystery; it is a monitored asset.

References and AI Use Transparency

- Dataset: NASA NeoWS (Near-Earth Object Web Service) API
- Tools: Python (pandas, scikit-learn, XGBoost, matplotlib)
- AI Assistance: ChatGPT and Gemini were used to help refine writing and structure. All analysis, modeling, and conclusions were completed and verified independently.